

Technical Assignment: AI Chatbot with RAG and MCP Tools

Objective

Build a working AI chatbot demonstrating orchestration between general conversation, RAG-based knowledge retrieval, MCP API tools, and multi-turn conversation context. The goal is to evaluate practical AI orchestration skills.

Area	Requirement
Conversation	Maintain context across multiple user messages
Orchestration	Decide whether to answer directly, use RAG, call a tool, or combine both
RAG	Retrieve answers from provided documents and show the source
MCP Tools	Connect to and execute business API tools
Input Handling	Ask for missing required information
Confirmation	Request confirmation before package changes
Error Handling	Handle retrieval and tool failures gracefully
Observability	Provide basic logs showing orchestration path

Required MCP Tools

Tool	Purpose
Get customer package	Return customer's current package
Get data balance	Return remaining mobile data
Create support ticket	Create ticket and return reference
Change customer package	Submit package change after confirmation

Mock data and mock APIs are acceptable.

Required Demo Scenarios

- Knowledge-base question answered through RAG.
- Tool request with missing information.
- Multi-turn package-change request requiring confirmation.
- Support-ticket creation request.
- Question not covered by the knowledge base.
- Simulated API or tool failure.

Deliverables

- Source code
- Architecture diagram
- Working demo (through local deployment from the code with READ.ME guidance or Cloud implementation link)
- Presentation

Architecture Diagram

Include the following major components:

- User Interface or API
- Chatbot Orchestration Layer
- LLM
- RAG Component and Vector Database
- MCP Client and MCP Tools
- Conversation Memory
- Business API or Mock Database
- Logging and Error Handling

It is not mandatory to call for a review meeting, but if requested below may be required:

1. Live Demonstrations

- Presentation.
- Demo execution.
- Questions

2. Live Change (Not mandatory)

During the interview, the interviewer may request a small change such as adding a new MCP tool, modifying a tool parameter, adding a knowledge-base document, changing a confirmation rule, or demonstrating a new orchestration scenario. The candidate should implement the change or clearly explain the required code modifications.

Assignment Duration

Estimated effort: 2 WDs.

Submission deadline: Within three calendar days.